

03-01-06-InfoSharingBinominal

```
source("../_common.R")
filtered_data <- readRDS("../artifacts/filtered_data.rds")
codebook <- readRDS("../artifacts/codebook.rds")
```

setup _____

libraries

```
library(glmTMB)
library(emmeans)
library(openxlsx)
library(modelsummary)
library(gtsummary)
library(broom)
library(DHARMA)
```

```
# Create workbook to save the analyses
wb <- createWorkbook()
excel_export_file_location <- "../artifacts/binomial_output.xlsx"

# Start fresh each time
if (file.exists(excel_export_file_location)) {
  file.remove(excel_export_file_location)
}
```

[1] TRUE

items key

Each of the items from the tasks are listed by the variable name. A key is created with the characteristics of each item.

```
item_vars <- c(
  "Q53_1", "Q53_2", "Q53_3", "Q53_4", "Q53_5", "Q53_6",
  "Q54_1", "Q54_2", "Q54_3", "Q54_4", "Q54_5", "Q54_6"
)

item_key <- tribble(
  ~ItemID, ~InfoImportance, ~InfoDistribution,
  "Q53_1", "High",          "Common",
  "Q53_4", "High",          "Common",
  "Q53_5", "High",          "Common",
  "Q53_2", "Low",           "Common",
  "Q53_3", "Low",           "Common",
  "Q53_6", "Low",           "Common",
  "Q54_2", "High",          "Unique",
  "Q54_3", "High",          "Unique",
  "Q54_6", "High",          "Unique",
  "Q54_1", "Low",           "Unique",
  "Q54_4", "Low",           "Unique",
  "Q54_5", "Low",           "Unique"
)
```

limit the data

Selects only the variables from the data set which will be used in the analysis.

```
dat_items <- filtered_data %>%
  select(
    ResponseId,

    # experimental predictors
    structure_number01,
    motivation_number01,

    # control variables
    Q14.1,
    sex,
    SES_mean,
```

```

    hispanic,
    NeedCog_Mean,
    DirtyDozen_Narcissism,
    DirtyDozen_Psychopathy,
    DirtyDozen_Machiavellianism,
    SV0_angle,

    # individual information items
    Q53_1,
    Q53_2,
    Q53_3,
    Q53_4,
    Q53_5,
    Q53_6,
    Q54_1,
    Q54_2,
    Q54_3,
    Q54_4,
    Q54_5,
    Q54_6
  ) %>%
  mutate(
    across(
      c(
        Q53_1, Q53_2, Q53_3,
        Q53_4, Q53_5, Q53_6,
        Q54_1, Q54_2, Q54_3,
        Q54_4, Q54_5, Q54_6
      ),
      ~ case_when(
        .x %in% c(1, 3) ~ 0,
        .x == 2 ~ 1,
        TRUE ~ NA_real_
      )
    )
  )
)

```

pivot

Data restructuring from wide to long.

```

dat_long <- dat_items %>%
  pivot_longer(
    cols = c(
      Q53_1, Q53_2, Q53_3,
      Q53_4, Q53_5, Q53_6,
      Q54_1, Q54_2, Q54_3,
      Q54_4, Q54_5, Q54_6
    ),
    names_to = "ItemID",
    values_to = "ItemResponse"
  )

```

update labels

Adds meaningful labels to the key variables for easier interpretation later.

```

# Combine key for items with data
# Update labels for variables
dat_long <- dat_long %>%
  left_join(item_key, by = "ItemID") %>%
  mutate(
    structure_number01 = factor(
      structure_number01,
      levels = c(0, 1),
      labels = c("Interindividual", "Intergroup")
    ),
    motivation_number01 = factor(
      motivation_number01,
      levels = c(0, 1),
      labels = c("Proself", "Prosocial")
    ),
    InfoDistribution = factor(
      InfoDistribution,
      levels = c("Common", "Unique")
    ),
    InfoImportance = factor(
      InfoImportance,
      levels = c("Low", "High"),
      labels = c("Low Importance", "High Importance")
    )
  )

```

checks

Are all of the items from the task present?

```
dat_long %>%
  distinct(
    ItemID,
    InfoImportance,
    InfoDistribution
  ) %>%
  arrange(ItemID)

# A tibble: 12 x 3
  ItemID InfoImportance InfoDistribution
  <chr>   <fct>             <fct>
1 Q53_1  High Importance    Common
2 Q53_2  Low Importance     Common
3 Q53_3  Low Importance     Common
4 Q53_4  High Importance    Common
5 Q53_5  High Importance    Common
6 Q53_6  Low Importance     Common
7 Q54_1  Low Importance     Unique
8 Q54_2  High Importance    Unique
9 Q54_3  High Importance    Unique
10 Q54_4  Low Importance     Unique
11 Q54_5  Low Importance     Unique
12 Q54_6  High Importance    Unique
```

Are there missing values?

```
dat_long %>%
  filter(
    is.na(InfoImportance) |
    is.na(InfoDistribution)
  ) %>%
  distinct(ItemID)
```

```
# A tibble: 0 x 1
# i 1 variable: ItemID <chr>
```

Counting

```
table(
  dat_long$InfoDistribution,
  dat_long$InfoImportance,
  useNA = "ifany"
)
```

	Low Importance	High Importance
Common	312	312
Unique	312	312

```
table(dat_long$ItemResponse, useNA = "ifany")
```

```

0    1
529 719
```

save data

```
# Export to excel file for sharing
write.xlsx(dat_long, "../artifacts/binominal_data.xlsx", asTable = TRUE, colWidths = "auto")
```

control model _____

create form

```
# This first part is only writing out the variables so that when I run the model it is easier
form_controls <- ItemResponse ~ # ItemResponse is if the participant chose share or not share
  InfoDistribution +
  InfoImportance +
  Q14.1 + # 14.1 is age
  sex + # These are control variables
  SES_mean +
  hispanic +
  NeedCog_Mean +
  DirtyDozen_Narcissism +
  DirtyDozen_Psychopathy +
```

```

DirtyDozen_Machiavellianism +
SVO_angle +
(1 | ResponseId) +
(1 | ItemID)

```

```

model_controls <- glmmTMB(
  form_controls,
  family = binomial(link = "logit"),
  data = dat_long
)

summary(model_controls)

```

```

Family: binomial ( logit )
Formula:
ItemResponse ~ InfoDistribution + InfoImportance + Q14.1 + sex +
  SES_mean + hispanic + NeedCog_Mean + DirtyDozen_Narcissism +
  DirtyDozen_Psychopathy + DirtyDozen_Machiavellianism + SVO_angle +
  (1 | ResponseId) + (1 | ItemID)
Data: dat_long

```

AIC	BIC	logLik	-2*log(L)	df.resid
1563.5	1635.2	-767.7	1535.5	1222

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
ResponseId	(Intercept)	0.4994	0.7067
ItemID	(Intercept)	0.1003	0.3167

Number of obs: 1236, groups: ResponseId, 103; ItemID, 12

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.3198464	1.2990450	1.016	0.30962
InfoDistributionUnique	-0.4379982	0.2233762	-1.961	0.04990 *
InfoImportanceHigh Importance	-0.8970107	0.2243309	-3.999	6.37e-05 ***
Q14.1	-0.0468183	0.0514629	-0.910	0.36295
sex	-0.5853947	0.2406769	-2.432	0.01500 *
SES_mean	0.0482443	0.0768867	0.627	0.53035
hispanic	-0.0530679	0.2059651	-0.258	0.79667
NeedCog_Mean	0.5462855	0.1711924	3.191	0.00142 **

DirtyDozen_Narcissism	-0.1502706	0.1341319	-1.120	0.26258
DirtyDozen_Psychopathy	-0.0990269	0.1445358	-0.685	0.49326
DirtyDozen_Machiavellianism	-0.1252753	0.1399786	-0.895	0.37081
SV0_angle	-0.0001403	0.0089992	-0.016	0.98757

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
confint(model_controls)
```

	2.5 %	97.5 %	Estimate
(Intercept)	-1.22623503	3.8659278668	1.3198464164
InfoDistributionUnique	-0.87580759	-0.0001888111	-0.4379982001
InfoImportanceHigh Importance	-1.33669124	-0.4573301877	-0.8970107154
Q14.1	-0.14768376	0.0540470683	-0.0468183462
sex	-1.05711284	-0.1136766312	-0.5853947357
SES_mean	-0.10245091	0.1989396003	0.0482443439
hispanic	-0.45675213	0.3506163616	-0.0530678829
NeedCog_Mean	0.21075452	0.8818165775	0.5462855463
DirtyDozen_Narcissism	-0.41316426	0.1126229774	-0.1502706415
DirtyDozen_Psychopathy	-0.38231193	0.1842581739	-0.0990268800
DirtyDozen_Machiavellianism	-0.39962829	0.1490777556	-0.1252752684
SV0_angle	-0.01777827	0.0174977618	-0.0001402561
Std.Dev.(Intercept) ResponseId	0.52998271	0.9422265651	0.7066567655
Std.Dev.(Intercept) ItemID	0.17110212	0.5861604658	0.3166911738

Only sex and need for cognition were significant. Therefore, only these two control variables are kept in the full model.

table

```
coef_table <- summary(model_controls)$coefficients$cond
ci <- confint(model_controls, parm = "beta_")

results <- data.frame(
  Predictor = rownames(coef_table),
  b = coef_table[, "Estimate"],
  SE = coef_table[, "Std. Error"],
  Wald_chisq = coef_table[, "z value"]^2,
  p = coef_table[, "Pr(>|z|)"],
  OR = exp(coef_table[, "Estimate"]),
  CI_lower = exp(ci[, 1]),
```

```

CI_upper = exp(ci[, 2]),
row.names = NULL
)

# Labels for dissertation table
predictor_labels <- c(
  "(Intercept)" = "Intercept",
  "InfoDistributionUnique" = "Common (0) vs. Unique (1)",
  "InfoImportanceLow Importance" = "Low (0) vs. High (1)",
  "Q14.1" = "Age",
  "sex" = "Sex (female = 1)",
  "SES_mean" = "Socioeconomic Status",
  "hispanic" = "Hispanic",
  "NeedCog_Mean" = "Need for Cognition",
  "DirtyDozen_Narcissism" = "Narcissism",
  "DirtyDozen_Psychopathy" = "Psychopathy",
  "DirtyDozen_Machiavellianism" = "Machiavellianism",
  "SVO_angle" = "SVO angle"
)

results$Predictor <- predictor_labels[results$Predictor]

# Add section rows
add_section <- function(label, template) {
  row <- template[1, ]
  row[] <- NA
  row$Predictor <- label
  row
}

results <- rbind(
  results,
  add_section("Within Participant Variables", results),
  add_section("Control Variables", results)
)

# Desired row order
row_order <- c(
  "Intercept",
  "Within Participant Variables",
  "Common (0) vs. Unique (1)",
  "Low (0) vs. High (1)",

```

```

"Control Variables",
"Age",
"Sex (female = 1)",
"Socioeconomic Status",
"Hispanic",
"Need for Cognition",
"Narcissism",
"Psychopathy",
"Machiavellianism",
"SVO angle"
)

final_results <- results[
  match(row_order, results$Predictor),
]

rownames(final_results) <- NULL

# Format regular statistics
stat_cols <- c("b", "SE", "Wald_chisq", "OR", "CI_lower", "CI_upper")

final_results[stat_cols] <-
  lapply(final_results[stat_cols], format_stat)

# Format p-values separately
final_results$p <-
  format_stat(final_results$p, p_value = TRUE)

# Combine OR and CI into one column
final_results$`OR [95% CI]` <- with(
  final_results,
  ifelse(
    OR == "",
    "",
    paste0(OR, " [", CI_lower, ", ", CI_upper, "]")
  )
)

# Remove separate OR and CI columns
final_results <- final_results[
  , !names(final_results) %in%
    c("OR", "CI_lower", "CI_upper")
]

```

```

]

controls_binomial <- final_results

rm(
  coef_table,
  ci,
  results,
  predictor_labels,
  row_order,
  stat_cols,
  final_results
)

controls_binomial

```

	Predictor	b	SE	Wald_chisq	p	OR	[95% CI]
1	Intercept	1.32	1.30	1.03	.31	3.74	[0.29, 47.75]
2	Within Participant Variables						
3	Common (0) vs. Unique (1)	-0.44	0.22	3.84	.05*	0.65	[0.42, 1.00]
4	<NA>						
5	Control Variables						
6	Age	-0.05	0.05	0.83	.36	0.95	[0.86, 1.06]
7	Sex (female = 1)	-0.59	0.24	5.92	.02*	0.56	[0.35, 0.89]
8	Socioeconomic Status	0.05	0.08	0.39	.53	1.05	[0.90, 1.22]
9	Hispanic	-0.05	0.21	0.07	.80	0.95	[0.63, 1.42]
10	Need for Cognition	0.55	0.17	10.18	.001**	1.73	[1.23, 2.42]
11	Narcissism	-0.15	0.13	1.26	.26	0.86	[0.66, 1.12]
12	Psychopathy	-0.10	0.14	0.47	.49	0.91	[0.68, 1.20]
13	Machiavellianism	-0.13	0.14	0.80	.37	0.88	[0.67, 1.16]
14	SV0 angle	-0.00	0.01	0.00	.99	1.00	[0.98, 1.02]

controls_binomial_table

```
save_output_to_workbook(wb, "Control Variables", controls_binomial)
```

full model _____

create form

```
# This first part is only writing out the variables so that when I run the model it is easier
form_binom <- ItemResponse ~ # ItemResponse is if the participant chose share or not share
  structure_number01 *
  motivation_number01 *
  InfoDistribution *
  InfoImportance +
  sex + # Only sex and NFC contributed to the model and were retained
  NeedCog_Mean +
  (1 | ResponseId) + # These are random intercepts to account for the participant's repeated
  (1 | ItemID)
```

model

```
# This second part is running the model
model_binom <- glmmTMB( # The package being used is glmmTMB
  form_binom, # This is the form that was created in the first step above
  family = binomial(link = "logit"), # Selects binomial distribution using logit link
  data = dat_long
)

summary(model_binom)
```

```
Family: binomial ( logit )
Formula:
ItemResponse ~ structure_number01 * motivation_number01 * InfoDistribution *
  InfoImportance + sex + NeedCog_Mean + (1 | ResponseId) + (1 | ItemID)
Data: dat_long
```

AIC	BIC	logLik	-2*log(L)	df.resid
1537.3	1639.9	-748.7	1497.3	1228

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
--------	------	----------	----------

```

ResponseId (Intercept) 0.4491    0.6701
ItemID      (Intercept) 0.1036    0.3219
Number of obs: 1248, groups: ResponseId, 104; ItemID, 12

```

Conditional model:

```

(Intercept)
structure_number01Intergroup
motivation_number01Prosocial
InfoDistributionUnique
InfoImportanceHigh Importance
sex
NeedCog_Mean
structure_number01Intergroup:motivation_number01Prosocial
structure_number01Intergroup:InfoDistributionUnique
motivation_number01Prosocial:InfoDistributionUnique
structure_number01Intergroup:InfoImportanceHigh Importance
motivation_number01Prosocial:InfoImportanceHigh Importance
InfoDistributionUnique:InfoImportanceHigh Importance
structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique
structure_number01Intergroup:motivation_number01Prosocial:InfoImportanceHigh Importance
structure_number01Intergroup:InfoDistributionUnique:InfoImportanceHigh Importance
motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceHigh Importance
structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceHigh Importance

(Intercept)
structure_number01Intergroup
motivation_number01Prosocial
InfoDistributionUnique
InfoImportanceHigh Importance
sex
NeedCog_Mean
structure_number01Intergroup:motivation_number01Prosocial
structure_number01Intergroup:InfoDistributionUnique
motivation_number01Prosocial:InfoDistributionUnique
structure_number01Intergroup:InfoImportanceHigh Importance
motivation_number01Prosocial:InfoImportanceHigh Importance
InfoDistributionUnique:InfoImportanceHigh Importance
structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique
structure_number01Intergroup:motivation_number01Prosocial:InfoImportanceHigh Importance
structure_number01Intergroup:InfoDistributionUnique:InfoImportanceHigh Importance
motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceHigh Importance
structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceHigh Importance

```

```

(Intercept)
structure_number01Intergroup
motivation_number01Prosocial
InfoDistributionUnique
InfoImportanceHigh Importance
sex
NeedCog_Mean
structure_number01Intergroup:motivation_number01Prosocial
structure_number01Intergroup:InfoDistributionUnique
motivation_number01Prosocial:InfoDistributionUnique
structure_number01Intergroup:InfoImportanceHigh Importance
motivation_number01Prosocial:InfoImportanceHigh Importance
InfoDistributionUnique:InfoImportanceHigh Importance
structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique
structure_number01Intergroup:motivation_number01Prosocial:InfoImportanceHigh Importance
structure_number01Intergroup:InfoDistributionUnique:InfoImportanceHigh Importance
motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceHigh Importance
structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceHigh Importance

```

```

(Intercept)
structure_number01Intergroup
motivation_number01Prosocial
InfoDistributionUnique
InfoImportanceHigh Importance
sex
NeedCog_Mean
structure_number01Intergroup:motivation_number01Prosocial
structure_number01Intergroup:InfoDistributionUnique
motivation_number01Prosocial:InfoDistributionUnique
structure_number01Intergroup:InfoImportanceHigh Importance
motivation_number01Prosocial:InfoImportanceHigh Importance
InfoDistributionUnique:InfoImportanceHigh Importance
structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique
structure_number01Intergroup:motivation_number01Prosocial:InfoImportanceHigh Importance
structure_number01Intergroup:InfoDistributionUnique:InfoImportanceHigh Importance
motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceHigh Importance
structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceHigh Importance

```

```

(Intercept)
structure_number01Intergroup
motivation_number01Prosocial
InfoDistributionUnique

```

```

InfoImportanceHigh Importance
sex
NeedCog_Mean
structure_number01Intergroup:motivation_number01Prosocial
structure_number01Intergroup:InfoDistributionUnique
motivation_number01Prosocial:InfoDistributionUnique
structure_number01Intergroup:InfoImportanceHigh Importance
motivation_number01Prosocial:InfoImportanceHigh Importance
InfoDistributionUnique:InfoImportanceHigh Importance
structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique
structure_number01Intergroup:motivation_number01Prosocial:InfoImportanceHigh Importance
structure_number01Intergroup:InfoDistributionUnique:InfoImportanceHigh Importance
motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceHigh Importance
structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceHigh Importance
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

# The model output shows estimates and random effects without an obvious convergence warning
# TRUE = passes convergence check
model_binom$sdr$pdHess

```

```
[1] TRUE
```

```

saveRDS(model_binom,
        file = file.path("../artifacts/model_binom.rds")
)

```

dispersion

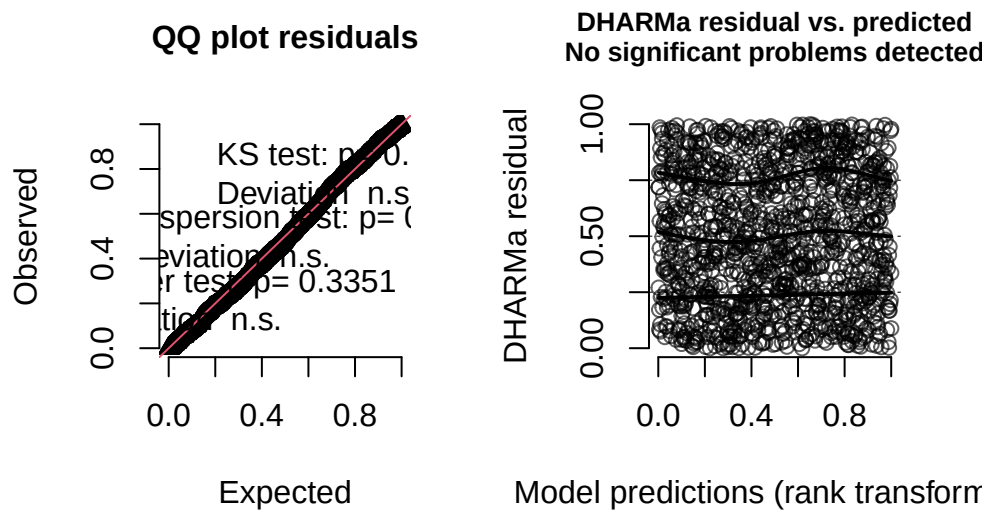
```

model_binom_sim_res <- simulateResiduals(
  fittedModel = model_binom
)

fig_model_binom_sim_res <- plot(model_binom_sim_res)

```

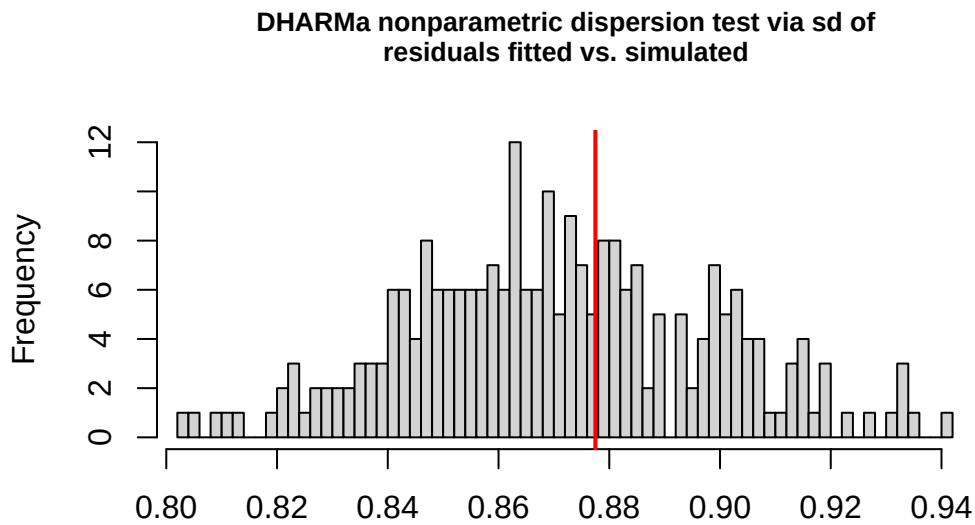
DHARMa residual



```
fig_model_binom_sim_res
```

NULL

```
# Test dispersion  
model_binom_dispersion <- testDispersion(  
  model_binom_sim_res  
)
```



Simulated values, red line = fitted model. p-value (two.sided) = 0.776

```
model_binom_dispersion
```

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 1.0081, p-value = 0.776
alternative hypothesis: two.sided
```

```
save_figure_to_workbook(
  wb = wb,
  sheet_name = "Binomial Res",
  figure_file = "../artifacts/fig_binomial_sim_res.png",
  plot_fun = function() plot(model_binom_sim_res),
  width = 10,
  height = 6,
  dpi = 300
)
```

Dispersion does not appear to be an issue due to non-significance in the test.

- **Dispersion test:** dispersion = 1.0081, $p = .776$, so there is no evidence of overdispersion.
- **KS test:** $p = .32279$, so the simulated residual distribution does not show a significant departure from what the model expects.
- **Outlier test:** $p = .3351$, so there is no evidence of an unusual number of outliers.
- The residual-versus-predicted panel explicitly says “**No significant problems detected.**”

table

```
coef_table <- summary(model_binom)$coefficients$cond
ci <- confint(model_binom, parm = "beta_")

results <- data.frame(
  Predictor = rownames(coef_table),
  b = coef_table[, "Estimate"],
  SE = coef_table[, "Std. Error"],
  Wald_chisq = coef_table[, "z value"]^2,
  p = coef_table[, "Pr(>|z|)"],
  OR = exp(coef_table[, "Estimate"]),
  CI_lower = exp(ci[, 1]),
  CI_upper = exp(ci[, 2]),
  row.names = NULL
)

predictor_labels <- c(
  "(Intercept)" = "Intercept",

  "sex" =
    "Sex (1 = female)",

  "NeedCog_Mean" =
    "Need for Cognition",

  "structure_number01Intergroup" =
    "Interindividual (0) vs. Intergroup (1)",

  "motivation_number01Prosocial" =
    "Proself (0) vs. Prosocial (1)",
```

```

"InfoDistributionUnique" =
  "Common (0) vs. Unique (1)",

"InfoImportanceLow Importance" =
  "Low (0) vs. High (1)",

"structure_number01Intergroup:motivation_number01Prosocial" =
  "Structure × Motivation",

"structure_number01Intergroup:InfoDistributionUnique" =
  "Structure × Distribution",

"motivation_number01Prosocial:InfoDistributionUnique" =
  "Motivation × Distribution",

"structure_number01Intergroup:InfoImportanceLow Importance" =
  "Structure × Importance",

"motivation_number01Prosocial:InfoImportanceLow Importance" =
  "Motivation × Importance",

"InfoDistributionUnique:InfoImportanceLow Importance" =
  "Distribution × Importance",

"structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique" =
  "Structure × Motivation × Distribution",

"structure_number01Intergroup:motivation_number01Prosocial:InfoImportanceLow Importance" =
  "Structure × Motivation × Importance",

"structure_number01Intergroup:InfoDistributionUnique:InfoImportanceLow Importance" =
  "Structure × Distribution × Importance",

"motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceLow Importance" =
  "Motivation × Distribution × Importance",

"structure_number01Intergroup:motivation_number01Prosocial:InfoDistributionUnique:InfoImportanceLow Importance" =
  "Structure × Motivation × Distribution × Importance"
)

results$Predictor <- predictor_labels[results$Predictor]

```

```

add_section <- function(label, template) {
  row <- template[1, ]
  row[] <- NA
  row$Predictor <- label
  row
}

results <- rbind(
  results,
  add_section("Control Variables", results),
  add_section("Between Manipulations", results),
  add_section("Within Manipulations", results),
  add_section("Interactions", results)
)

row_order <- c(
  "Intercept",

  "Control Variables",
  "Sex (1 = female)",
  "Need for Cognition",

  "Between Manipulations",
  "Interindividual (0) vs. Intergroup (1)",
  "Proself (0) vs. Prosocial (1)",

  "Within Manipulations",
  "Common (0) vs. Unique (1)",
  "Low (0) vs. High (1)",

  "Interactions",
  "Structure × Motivation",
  "Structure × Distribution",
  "Motivation × Distribution",
  "Structure × Importance",
  "Motivation × Importance",
  "Distribution × Importance",
  "Structure × Motivation × Distribution",
  "Structure × Motivation × Importance",
  "Structure × Distribution × Importance",
  "Motivation × Distribution × Importance",
  "Structure × Motivation × Distribution × Importance"
)

```

```

)

final_model_results <- results[
  match(row_order, results$Predictor),
]

rownames(final_model_results) <- NULL

stat_cols <- c(
  "b",
  "SE",
  "Wald_chisq",
  "OR",
  "CI_lower",
  "CI_upper"
)

final_model_results[stat_cols] <-
  lapply(final_model_results[stat_cols], format_stat)

final_model_results$p <-
  format_stat(final_model_results$p, p_value = TRUE)

final_model_results$`OR [95% CI]` <- with(
  final_model_results,
  ifelse(
    OR == "",
    "",
    paste0(OR, " [", CI_lower, ", ", CI_upper, "]")
  )
)

final_model_results <- final_model_results[
  , !names(final_model_results) %in%
    c("OR", "CI_lower", "CI_upper")
]

model_binomial_table <- final_model_results
model_binomial_table

```

	Predictor	b	SE	Wald_chisq	p
1	Intercept	-0.67	0.67	1.00	.32

2	Control Variables				
3	Sex (1 = female)	-0.58 0.21	7.44	.01**	
4	Need for Cognition	0.64 0.16	15.46	< .001**	
5	Between Manipulations				
6	Interindividual (0) vs. Intergroup (1)	-0.15 0.43	0.12	.73	
7	Proself (0) vs. Prosocial (1)	-0.08 0.42	0.04	.85	
8	Within Manipulations				
9	Common (0) vs. Unique (1)	-0.93 0.46	4.17	.04*	
10	<NA>				
11	Interactions				
12	Structure × Motivation	0.97 0.61	2.55	.11	
13	Structure × Distribution	0.66 0.53	1.53	.22	
14	Motivation × Distribution	1.00 0.52	3.75	.05	
15	<NA>				
16	<NA>				
17	<NA>				
18	Structure × Motivation × Distribution	-1.69 0.75	5.15	.02*	
19	<NA>				
20	<NA>				
21	<NA>				
22	<NA>				
	OR [95% CI]				
1	0.51 [0.14, 1.91]				
2					
3	0.56 [0.37, 0.85]				
4	1.89 [1.38, 2.60]				
5					
6	0.86 [0.37, 2.01]				
7	0.92 [0.41, 2.10]				
8					
9	0.39 [0.16, 0.96]				
10					
11					
12	2.64 [0.80, 8.70]				
13	1.93 [0.68, 5.43]				
14	2.73 [0.99, 7.53]				
15					
16					
17					
18	0.18 [0.04, 0.79]				
19					
20					
21					

```
model_binomial_table
```

```
save_output_to_workbook(wb, "Binomial Table", model_binomial_table)
```

omnibus test

overall main effects and interactions

similar to Type III ANOVA-style table

```
joint_results <- joint_tests(model_binom)
```

```
joint_results
```

model	term	df1	df2
	structure_number01	1	Inf
	motivation_number01	1	Inf
	InfoDistribution	1	Inf
	InfoImportance	1	Inf
	sex	1	Inf
	NeedCog_Mean	1	Inf
	structure_number01:motivation_number01	1	Inf
	structure_number01:InfoDistribution	1	Inf
	structure_number01:InfoImportance	1	Inf
	motivation_number01:InfoDistribution	1	Inf
	motivation_number01:InfoImportance	1	Inf
	InfoDistribution:InfoImportance	1	Inf
	structure_number01:motivation_number01:InfoDistribution	1	Inf
	structure_number01:motivation_number01:InfoImportance	1	Inf
	structure_number01:InfoDistribution:InfoImportance	1	Inf
	motivation_number01:InfoDistribution:InfoImportance	1	Inf
	structure_number01:motivation_number01:InfoDistribution:InfoImportance	1	Inf
F.ratio	Chisq	p.value	
0.296	0.296	0.5861	
15.855	15.855	0.0001	
3.826	3.826	0.0505	
16.367	16.367	0.0001	
7.439	7.439	0.0064	
15.457	15.457	0.0001	

4.124	4.124	0.0423
3.549	3.549	0.0596
6.802	6.802	0.0091
0.059	0.059	0.8084
4.275	4.275	0.0387
0.127	0.127	0.7216
0.058	0.058	0.8093
11.392	11.392	0.0007
1.327	1.327	0.2494
0.130	0.130	0.7182
12.112	12.112	0.0005

decomposition by info I & D

```
joint_panels <- joint_tests(
  model_binom,
  by = c("InfoImportance", "InfoDistribution")
)

joint_panels
```

InfoImportance = Low Importance, InfoDistribution = Common:

model term	df1	df2	F.ratio	Chisq	p.value
structure_number01	1	Inf	1.204	1.204	0.2725
motivation_number01	1	Inf	1.761	1.761	0.1845
sex	1	Inf	7.439	7.439	0.0064
NeedCog_Mean	1	Inf	15.457	15.457	0.0001
structure_number01:motivation_number01	1	Inf	2.550	2.550	0.1103

InfoImportance = High Importance, InfoDistribution = Common:

model term	df1	df2	F.ratio	Chisq	p.value
structure_number01	1	Inf	0.027	0.027	0.8694
motivation_number01	1	Inf	12.545	12.545	0.0004
sex	1	Inf	7.439	7.439	0.0064
NeedCog_Mean	1	Inf	15.457	15.457	0.0001
structure_number01:motivation_number01	1	Inf	19.953	19.953	<.0001

InfoImportance = Low Importance, InfoDistribution = Unique:

model term	df1	df2	F.ratio	Chisq	p.value
structure_number01	1	Inf	0.249	0.249	0.6175
motivation_number01	1	Inf	3.795	3.795	0.0514

sex	1	Inf	7.439	7.439	0.0064
NeedCog_Mean	1	Inf	15.457	15.457	0.0001
structure_number01:motivation_number01	1	Inf	1.575	1.575	0.2095

InfoImportance = High Importance, InfoDistribution = Unique:

model term	df1	df2	F.ratio	Chisq	p.value
structure_number01	1	Inf	8.357	8.357	0.0038
motivation_number01	1	Inf	11.951	11.951	0.0005
sex	1	Inf	7.439	7.439	0.0064
NeedCog_Mean	1	Inf	15.457	15.457	0.0001
structure_number01:motivation_number01	1	Inf	1.345	1.345	0.2462

Information condition	Structure	Motivation	Structure × Motivation	What it means
High, Common	.869	< .001	< .001	The effect of structure depends on motivation. Unpack with simple interactions.
High, Unique	.004	< .001	.246	Structure and motivation both have significant effects, with <u>no</u> evidence that their effects depend on one another.
Low, Common	.273	.185	.110	<u>No</u> significant effects of structure, motivation, or their interaction.

Information condition	Structure	Motivation	Structure × Motivation	What it means
Low, Unique	.618	.051	.210	No significant effects at the conventional .05 level. Motivation is close, but does not reach significance.

```
save_output_to_workbook(
  wb,
  sheet_name = "Joint Panels",
  output = joint_panels
)
```

emmeans() —————

<https://rvlenth.github.io/emmeans/>

The purpose of this analysis is to break down the four-way interaction to understand what is happening with each of the variables.

Because **emmeans** with logistic models often reports contrasts as **odds ratios**, this will usually pair with the predicted probabilities from the bar chart.

These are means that are from the model, which means that they have all of the other predictor variables that were included in the model as well.

Also, random note about the package: the model was built with glmmTMB, so emmeans is relying on methods supplied by glmmTMB to tell it how to use the model data, fixed effects, covariance matrix, etc to compute the estimated marginal means.

<https://rdr.io/r/stats/coef.html>

coef() is a generic function which extracts model coefficients from objects returned by modeling functions. **coefficients** is an alias for it.

pairs() is used on emmeans objects to probe significant effects or interactions

s * m

Estimated marginal means for Structure x Motivation across combinations of Distribution x Importance.

```
emm_sm <- emmeans(  
  model_binom,  
  ~ structure_number01 * motivation_number01 |  
    InfoDistribution * InfoImportance,  
  type = "response"  
)
```

pairs()

Six pairwise comparisons within each information combination.

Limited interpretation

```
pairs(  
  emm_sm,  
  adjust = "holm"  
)
```

InfoDistribution = Common, InfoImportance = Low Importance:

contrast	odds.ratio	SE	df	null
Interindividual Proself / Intergroup Proself	1.1631	0.5030	Inf	1
Interindividual Proself / Interindividual Prosocial	1.0843	0.4540	Inf	1
Interindividual Proself / Intergroup Prosocial	0.4774	0.2130	Inf	1
Intergroup Proself / Interindividual Prosocial	0.9323	0.3880	Inf	1
Intergroup Proself / Intergroup Prosocial	0.4104	0.1820	Inf	1
Interindividual Prosocial / Intergroup Prosocial	0.4402	0.1890	Inf	1

z.ratio	p.value
0.350	1.0000
0.193	1.0000
-1.657	0.3901
-0.169	1.0000
-2.012	0.2651
-1.910	0.2804

InfoDistribution = Unique, InfoImportance = Low Importance:

contrast	odds.ratio	SE	df	null
Interindividual Proself / Intergroup Proself	0.6040	0.2490	Inf	1

Interindividual Proself / Interindividual Prosocial	0.3976	0.1620	Inf	1
Interindividual Proself / Intergroup Prosocial	0.4937	0.1980	Inf	1
Intergroup Proself / Interindividual Prosocial	0.6582	0.2720	Inf	1
Intergroup Proself / Intergroup Prosocial	0.8174	0.3330	Inf	1
Interindividual Prosocial / Intergroup Prosocial	1.2419	0.4980	Inf	1

z.ratio p.value

-1.222	0.8869
-2.270	0.1392
-1.757	0.3946
-1.012	0.9348
-0.494	1.0000
0.541	1.0000

InfoDistribution = Common, InfoImportance = High Importance:

contrast	odds.ratio	SE	df	null
Interindividual Proself / Intergroup Proself	0.2839	0.1190	Inf	1
Interindividual Proself / Interindividual Prosocial	0.0956	0.0411	Inf	1
Interindividual Proself / Intergroup Prosocial	0.3707	0.1520	Inf	1
Intergroup Proself / Interindividual Prosocial	0.3368	0.1410	Inf	1
Intergroup Proself / Intergroup Prosocial	1.3059	0.5200	Inf	1
Interindividual Prosocial / Intergroup Prosocial	3.8773	1.5700	Inf	1

z.ratio p.value

-2.997	0.0109
-5.458	<.0001
-2.427	0.0305
-2.597	0.0282
0.670	0.5030
3.337	0.0042

InfoDistribution = Unique, InfoImportance = High Importance:

contrast	odds.ratio	SE	df	null
Interindividual Proself / Intergroup Proself	1.6553	0.7010	Inf	1
Interindividual Proself / Interindividual Prosocial	0.2606	0.1050	Inf	1
Interindividual Proself / Intergroup Prosocial	0.8435	0.3370	Inf	1
Intergroup Proself / Interindividual Prosocial	0.1574	0.0666	Inf	1
Intergroup Proself / Intergroup Prosocial	0.5096	0.2130	Inf	1
Interindividual Prosocial / Intergroup Prosocial	3.2365	1.2800	Inf	1

z.ratio p.value

1.190	0.4681
-3.325	0.0044
-0.425	0.6705
-4.368	0.0001
-1.615	0.3186

2.970 0.0119

Results are averaged over the levels of: sex
P value adjustment: holm method for 6 tests
Tests are performed on the log odds ratio scale

contrasts_emm_sm

```
save_output_to_workbook(  
  wb,  
  sheet_name = "SM Pairs",  
  output =  
    pairs(  
      emm_sm,  
      adjust = "holm"  
    )  
)
```

fig_emm_sm

```
emm_sm_df <- as.data.frame(emm_sm)
```

```
save_output_to_workbook(wb, "SM emmean", emm_sm_df)
```

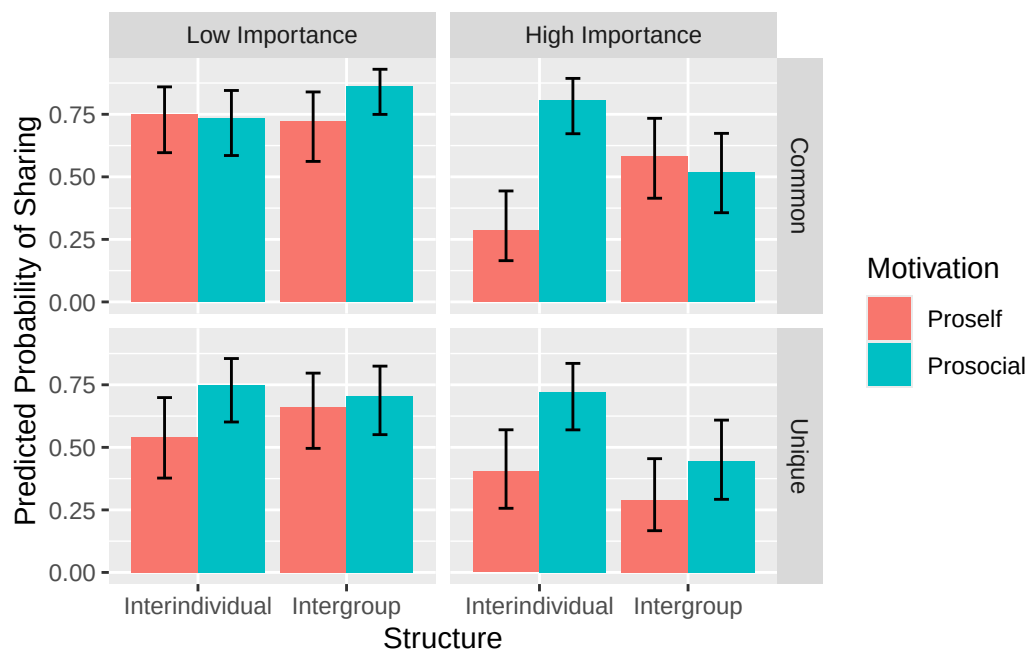
```
fig_emm_sm <- ggplot(  
  emm_sm_df,  
  aes(  
    x = structure_number01,  
    y = prob,  
    fill = motivation_number01  
  )  
) +  
  geom_col(  
    position = position_dodge(width = 0.9)  
  ) +  
  geom_errorbar(  
    aes(  
      ymin = asymp.LCL,  
      ymax = asymp.UCL  
    ),  
    width = 0.2,
```

```

    position = position_dodge(width = 0.9)
  ) +
  facet_grid(
    InfoDistribution ~ InfoImportance
  ) +
  labs(
    x = "Structure",
    y = "Predicted Probability of Sharing",
    fill = "Motivation"
  )
)

```

fig_emm_sm



fig_emm_sm

```

save_figure_to_workbook(
  wb = wb,
  sheet_name = "SM Figure",
  figure_file = "../artifacts/SM_Interaction.png",
  plot_obj = fig_emm_sm,
  width = 8,
  height = 6,
)

```

```

    dpi = 300
  )

```

significant panels ---

Explore the significant panels

- High/Common
- High/Unique

high & common

best both sets of simple effects because both Structure x Motivation is significant

```

emm_high_common <- emmeans(
  model_binom,
  ~ structure_number01 * motivation_number01, # note *
  at = list(
    InfoImportance = "High Importance",
    InfoDistribution = "Common"
  ),
  type = "response"
)

```

NOTE: Results may be misleading due to involvement in interactions

```
emm_high_common
```

structure_number01	motivation_number01	prob	SE	df	asympt.LCL	asympt.UCL
Interindividual	Proself	0.284	0.0724	Inf	0.165	0.444
Intergroup	Proself	0.583	0.0844	Inf	0.414	0.734
Interindividual	Prosocial	0.806	0.0562	Inf	0.672	0.893
Intergroup	Prosocial	0.517	0.0838	Inf	0.357	0.674

Results are averaged over the levels of: sex

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

compare Structure within each Motivation

```
hc_structure <- emmeans(  
  model_binom,  
  ~ structure_number01 | motivation_number01, # note order and |  
  at = list(  
    InfoImportance = "High Importance",  
    InfoDistribution = "Common"  
  ),  
  type = "response"  
)
```

NOTE: Results may be misleading due to involvement in interactions

```
pairs(hc_structure)
```

motivation_number01 = Proself:

contrast	odds.ratio	SE	df	null	z.ratio	p.value
Interindividual / Intergroup	0.284	0.119	Inf	1	-2.997	0.0027

motivation_number01 = Prosocial:

contrast	odds.ratio	SE	df	null	z.ratio	p.value
Interindividual / Intergroup	3.877	1.570	Inf	1	3.337	0.0008

Results are averaged over the levels of: sex

Tests are performed on the log odds ratio scale

compare Motivation within each Structure

```
hc_motivation <- emmeans(  
  model_binom,  
  ~ motivation_number01 | structure_number01, # note order and |  
  at = list(  
    InfoImportance = "High Importance",  
    InfoDistribution = "Common"  
  ),  
  type = "response"  
)
```

NOTE: Results may be misleading due to involvement in interactions

```
pairs(hc_motivation)
```

```
structure_number01 = Interindividual:
  contrast      odds.ratio      SE  df null z.ratio p.value
Proself / Prosocial    0.0956 0.0411 Inf   1  -5.458  <.0001
```

```
structure_number01 = Intergroup:
  contrast      odds.ratio      SE  df null z.ratio p.value
Proself / Prosocial    1.3059 0.5200 Inf   1   0.670  0.5030
```

Results are averaged over the levels of: sex
Tests are performed on the log odds ratio scale

high & unique

no significant structure x motivation interaction, but reason to look more closely at the prosocial/intergroup

starting with the four predicted probabilities

```
emm_high_unique <- emmeans(
  model_binom,
  ~ structure_number01 * motivation_number01, # note *
  at = list(
    InfoImportance = "High Importance",
    InfoDistribution = "Unique"
  ),
  type = "response"
)
```

NOTE: Results may be misleading due to involvement in interactions

```
emm_high_unique
```

structure_number01	motivation_number01	prob	SE	df	asympt.LCL	asympt.UCL
Interindividual	Proself	0.403	0.0828	Inf	0.256	0.570
Intergroup	Proself	0.290	0.0750	Inf	0.167	0.455
Interindividual	Prosocial	0.722	0.0689	Inf	0.570	0.836
Intergroup	Prosocial	0.445	0.0836	Inf	0.292	0.609

Results are averaged over the levels of: sex
 Confidence level used: 0.95
 Intervals are back-transformed from the logit scale

Are those two structure differences themselves significantly different from one another? No, there is insufficient evidence that the structure effect is statistically different between Proself and Prosocial. Shown in the contrast below.

```
pairs(
  emm_high_unique,
  adjust = "holm"
)
```

contrast	odds.ratio	SE	df	null
Interindividual Proself / Intergroup Proself	1.655	0.7010	Inf	1
Interindividual Proself / Interindividual Prosocial	0.261	0.1050	Inf	1
Interindividual Proself / Intergroup Prosocial	0.844	0.3370	Inf	1
Intergroup Proself / Interindividual Prosocial	0.157	0.0666	Inf	1
Intergroup Proself / Intergroup Prosocial	0.510	0.2130	Inf	1
Interindividual Prosocial / Intergroup Prosocial	3.237	1.2800	Inf	1

z.ratio	p.value
1.190	0.4681
-3.325	0.0044
-0.425	0.6705
-4.368	0.0001
-1.615	0.3186
2.970	0.0119

Results are averaged over the levels of: sex
 P value adjustment: holm method for 6 tests
 Tests are performed on the log odds ratio scale

Structure x Motivation

Among Prosocial participants, is sharing lower in the Intergroup condition than in the Interindividual condition?

```
hu_structure <- emmeans(
  model_binom,
  ~ structure_number01 | motivation_number01,
  at = list(
```

```

      InfoImportance = "High Importance",
      InfoDistribution = "Unique"
    ),
    type = "response"
  )

```

NOTE: Results may be misleading due to involvement in interactions

```
pairs(hu_structure)
```

```

motivation_number01 = Proself:
  contrast          odds.ratio    SE  df null z.ratio p.value
Interindividual / Intergroup      1.66 0.701 Inf    1   1.190  0.2340

```

```

motivation_number01 = Prosocial:
  contrast          odds.ratio    SE  df null z.ratio p.value
Interindividual / Intergroup      3.24 1.280 Inf    1   2.970  0.0030

```

Results are averaged over the levels of: sex
 Tests are performed on the log odds ratio scale

Motivation within Structure

Within the Intergroup condition, do Prosocial participants differ from Proself participants?

```

hu_motivation <- emmeans(
  model_binom,
  ~ motivation_number01 | structure_number01,
  at = list(
    InfoImportance = "High Importance",
    InfoDistribution = "Unique"
  ),
  type = "response"
)

```

NOTE: Results may be misleading due to involvement in interactions

```
pairs(hu_motivation)
```

```
structure_number01 = Interindividual:
  contrast          odds.ratio    SE  df null z.ratio p.value
Proself / Prosocial    0.261 0.105 Inf   1  -3.325  0.0009
```

```
structure_number01 = Intergroup:
  contrast          odds.ratio    SE  df null z.ratio p.value
Proself / Prosocial    0.510 0.213 Inf   1  -1.615  0.1062
```

Results are averaged over the levels of: sex
 Tests are performed on the log odds ratio scale

high I & unique & structure

Among prosocial participants, is the intergroup different from the interindividual? (And also, among the proself participants, is the intergroup different from the interindividual?)

Yes, there is a significant different among Prosocial participants. (simple effect of structure?)

Among prosocial participants, the predicted probability of sharing high-importance, unique information was significantly higher in the interindividual condition (.722) than in the intergroup condition (.445), OR = 3.24, $p = .003$.

```
emm_high_unique_structure <- emmeans(
  model_binom,
  ~ structure_number01 |
    motivation_number01,
  at = list(
    InfoImportance = "High Importance",
    InfoDistribution = "Unique"
  ),
  type = "response"
)
```

NOTE: Results may be misleading due to involvement in interactions

```
pairs(
  emm_high_unique_structure,
  adjust = "holm"
)
```

```
motivation_number01 = Proself:
  contrast          odds.ratio    SE  df null z.ratio p.value
```

Interindividual / Intergroup	1.66	0.701	Inf	1	1.190	0.2340
------------------------------	------	-------	-----	---	-------	--------

motivation_number01 = Prosocial:

contrast	odds.ratio	SE	df	null	z.ratio	p.value
Interindividual / Intergroup	3.24	1.280	Inf	1	2.970	0.0030

Results are averaged over the levels of: sex

Tests are performed on the log odds ratio scale

HU Prosocial emmean

```
save_output_to_workbook(
  wb,
  "HU Prosocial emmean",
  emm_high_unique_structure
)
```

HU Prosocial Contrast

```
save_output_to_workbook(
  wb,
  sheet_name = "HU Prosocial Contrast",
  output = pairs(
    emm_high_unique_structure,
    adjust = "holm"
  )
)
```

Save to excel _____

```
saveWorkbook(
  wb,
  excel_export_file_location,
  overwrite = TRUE
)
```

End _____

```
print("end of script")
```

```
[1] "end of script"
```